

JAEWOOK LEE

Curriculum Vitae

Personal Information

ADDRESS Packard 243,
350 Jane Stanford Way,
Stanford, CA 94305

PHONE +82-10-3539-1857
EMAIL jwl99 (at) stanford.edu
WEBSITE jwlee-primal.com

Education

CURRENT	Stanford University Ph.D. in Electrical Engineering	Stanford, CA, United States
FEB 2025	Korea Advanced Institute of Science and Technology M.S. in Artificial Intelligence (Advisor: Prof. Chulhee Yun) GPA: 4.25/4.3	Seoul, South Korea
FEB 2023	Korea Advanced Institute of Science and Technology B.S. in Electrical Engineering & Mathematical Sciences (<i>Double Major</i>) GPA: 4.07/4.3, SUMMA CUM LAUDE <i>Graduated with Excellence in Leadership and Volunteer Activity</i>	Daejeon, South Korea
FEB 2018	Sejong Science High School	Seoul, South Korea

Research Interests

I am interested in **optimization theory** and **statistics**. This includes broad topics of convex/nonconvex/stochastic optimization, statistical learning theory, sampling, and applications to practical problem settings in AI/ML. Recently, I have been particularly working on the following topics:

- Automated proofs and parameter search for convex optimization algorithms
- Performative prediction under endogenous and exogenous distribution shifts
- Monotone operators and applications to optimization theory
- Nonconvex, nonsmooth optimization of matrix-input functions
- Block coordinate algorithms and multi-player games
- Wasserstein gradient flows and applications to statistics

Publications

- [1] **Jaewook Lee**, Ernest K. Ryu, Chulhee Yun. Nesterov Acceleration with Operator Decomposition. *Preparing for submission*.
- [2] Donghwa Kim, **Jaewook Lee**, Chulhee Yun. Provable Benefit of Random Permutations over Uniform Sampling in Stochastic Coordinate Descent. *International Conference on Machine Learning (ICML)*, 2025.
- [3] **Jaewook Lee**^{*}, Hanseul Cho^{*}, Chulhee Yun. Fundamental Benefit of Alternating Updates in Minimax Optimization. *International Conference on Machine Learning (ICML)*, 2024. **Spotlight**.
- [4] Jaeyoung Cha, **Jaewook Lee**, Chulhee Yun. Tighter Lower Bounds for Shuffling SGD: Random Permutations and Beyond. *International Conference on Machine Learning (ICML)*, 2023. **Oral**.

^{*}Equal Contribution.

Projects

National Institute of Environmental Research (NIER)
Government Research Project

AUG 2022 – DEC 2024

- Real-time ozone concentration prediction (Time-series prediction of chemical & meteorological data)

Talks

- AUG 22ND, 2024 **2024 Digital Innovators' Symposium**, Seoul, South Korea ([page](#))
Title: Fundamental Benefit of Alternating Updates in Minimax Optimization
- AUG 13TH, 2024 **SNU-KAIST AI/ML Theory Workshop**, Gangneung, South Korea ([page](#))
Title: Exploiting Coordinate Structures in Optimization Algorithms

Awards & Honors

- 2024 ICML, **Spotlight Paper**
- 2023 ICML, **Oral Presentation**
- FALL 2021 KAIST MAS, **PoW (Math Problem of the Week)** - 3rd Prize
Math problem-solving competition for the Fall semester
- SPRING 2021 KAIST EE, **Dean's List Award**
Awarded to the top 3% of all EE students
- FALL 2020 KAIST EE, **Academic Excellence Scholarship**
Awarded to the top 4 students in EE
- FALL 2020 KAIST EE, **Dean's List Award**
- FALL 2019 KAIST EE, **Dean's List Award**
- FALL 2018 KAIST, **Freshman Dean's List Award**
Awarded to the top 3% of all freshman students

Professional Services

Journal/Conference Reviewer. ICLR (2025-26), ICML (2025), NeurIPS (2024-25), JMLR (2024)

Teaching Experience

KAIST Freshman Tutoring Program. Calculus II (2020 Fall, 2022 Spring)
General Physics I (2019 Spring)

Skills

Languages. English (*Highly Proficient*), Korean (*Native*)
TOEFL 115/120 (R29/L30/S29/W27), TOEIC 985/990

Computer Languages. Python (PyTorch Libraries), MATLAB, Julia, $\text{\LaTeX}$

Extracurricular Activities

- 2025 **Project Starlight Orchestra** First Violinist
- 2023 **Merry Orchestra, Original** First Violinist
- 2018-2022 **KAIST Orchestra** First Violinist (2019 Concertmaster)
- 2020-2022 **KAIST CGC** (*Communication Globalization Committee*), English Translator
- 2018-2021 **KAIST EDGE** (*Table Tennis Club*) Member
- 2020-2021 **KAIST UA** (*Undergrad. Assoc.*) Bureau of Welfare & Bureau of International Affairs
- 2020-2021 **KAIST FEEL** (*EE Conference Camp*, [link](#)) Program Director
- 2021 **KAIST 50th Anniversary Conference "Pioneers: 2071"**
Program Director (Scenario Author & Assistant MC)
- 2018-2019 **KAIST FSC** (*Freshman Student Council*)